EXPERIMENT 21: Prolog Program to Implement Towers of Hanoi

Aim

To write and execute a Prolog program to implement the **Towers of Hanoi** problem using recursion.

Algorithm

1. Start the Prolog program.
2. Define a predicate hanoi(N, Source, Destination, Auxiliary).
3. For one disk, move the disk directly from the source peg to the destination peg.
4. For more than one disk, recursively move N−1 disks from source to auxiliary.
5. Move the largest disk from source to destination.
6. Recursively move the N−1 disks from auxiliary to destination.
7. Display each disk movement.
8. Stop when all disks are moved to the destination peg.

Program Code:

```
% Towers of Hanoi


hanoi(1, Source, Destination, _) :-
   write('Move disk from '),
   write(Source),
   write(' to '),
   write(Destination),
   nl.


hanoi(N, Source, Destination, Auxiliary) :-
   N > 1,
   M is N - 1,
   hanoi(M, Source, Auxiliary, Destination),
```

hanoi(1, Source, Destination, Auxiliary),

hanoi(M, Auxiliary, Destination, Source).

Output:

```
(1656 ms) yes
| ?- consult('C:/Users/Hp/Documents/ex-21.pl').
compiling C:/Users/Hp/Documents/ex-21.pl for byte code...
C:/Users/Hp/Documents/ex-21.pl compiled, 14 lines read - 1392 bytes written, 14 ms

yes
| ?- hanoi(3, left, right, middle).
Move disk from left to right
Move disk from left to middle
Move disk from right to middle
Move disk from left to right
Move disk from middle to left
Move disk from middle to right
Move disk from left to right

true ?
```

Result

The Prolog program for the Towers of Hanoi problem was successfully implemented using recursion. The required sequence of disk movements was generated correctly.